Signals & Systems

Detailed Solutions

GATE INSTRUMENTATION ENGINEERING

PREPFUSION

Previous Year Questions Solutions

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Basics of Signals

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Q1.26.1 NAT 2026 2M Ans: 2.5

Given:

$$f(t) = \begin{cases} 1, & t \in [0, 2] \\ -t + 3, & t \in [2, 3] \\ 0, & \text{otherwise} \end{cases}$$

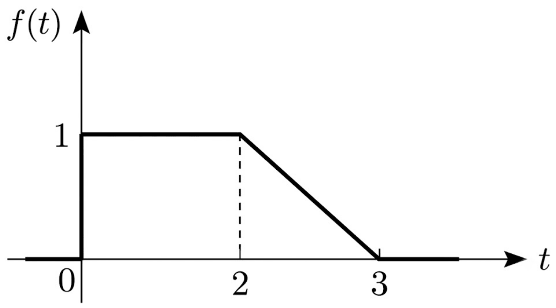

Fig. Solution Q1.26.1

$$\int_{-\infty}^{\infty} f(t)\, dt = 2.5$$

$$\text{Area} = 2 + \frac{1}{2} \times 1 \times 1 = 2.5$$

Hence, the correct answer is **2.5**.

$$\boxed{2.5}$$

Q1.26.2 MCQ 2026 2M Ans: CLICK FOR VIDEO SOLUTION

We need to determine

$$y(t) = 1 - x(4 - t)$$

The transformations are applied in the following order:

(i) $x(t) \to x(t + 4)$

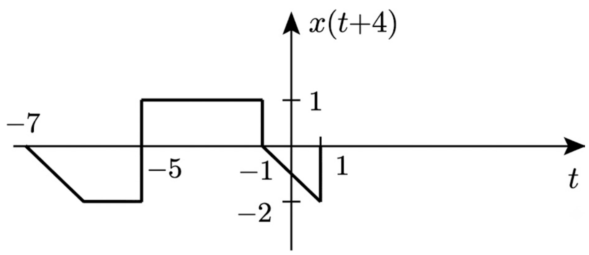

Fig. Solution Q1.26.2

(ii) $x(-t + 4)$

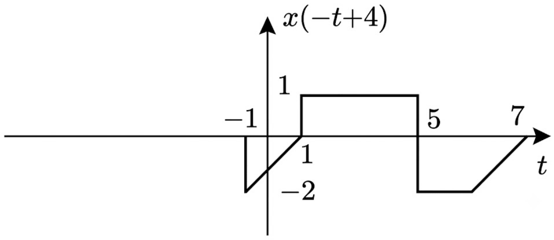

Fig. Solution Q1.26.2

(iii) $y(t) = 1 - x(-t+4)$

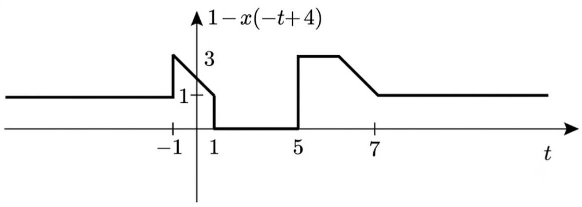

Fig. Solution Q1.26.2

Key Points

- When applying continuous-time transformations to $x(at+b)$, time-shifting ($t \to t+b$) should be performed before time-reversal/scaling.
- Amplitude transformations like $1 - f(t)$ invert the signal vertically across the horizontal axis before adding the DC shift.

Q1.26.3 MCQ 2026 1M Ans: D ● ○ ○

Given:

$$x(t) = 4\sin(15\pi t) + 7\cos(4\pi t)$$

For $x_1(t)$, $\omega_1 = 15\pi$,

$$T_1 = \frac{2\pi}{15\pi} = \frac{2}{15}$$

For $x_2(t)$, $\omega_2 = 4\pi$,

$$T_2 = \frac{2\pi}{4\pi} = \frac{1}{2}$$

The sum of periodic signals is periodic if the ratio of their periods is rational.

$$\frac{T_1}{T_2} = \frac{2/15}{1/2} = \frac{4}{15}$$

This is rational, so the signal is periodic.

$$T_0 = \text{LCM}[T_1, T_2]$$

$$T_0 = \text{LCM}\left[\frac{2}{15}, \frac{1}{2}\right]$$

$$T_0 = \frac{\text{LCM}(2,1)}{\text{HCF}(15,2)} = \frac{2}{1}$$

$$T_0 = 2 \text{ seconds}$$

Hence, the correct option is **(D)**.

$$\boxed{\text{(D) } 2}$$

Q1.25.1 MCQ 2025 1M Ans: B ● ○ ○

Given: $x(t) = e^{j9t} + e^{j5t}$ where $j = \sqrt{-1}$,

$$|x(t)| = \left|e^{j9t} + e^{j5t}\right|$$

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$x(t) = \cos 9t + j\sin 9t + \cos 5t + j\sin 5t$$

$$|x(t)| = \sqrt{(\cos 5t + \cos 9t)^2 + (\sin 5t + \sin 9t)^2}$$

$$|x(t)| = \sqrt{\cos^2 5t + \cos^2 9t + 2\cos 5t\cos 9t + \sin^2 5t + \sin^2 9t + 2\sin 5t\sin 9t}$$

$$|x(t)| = \sqrt{(1 + 1 + 2\cos 5t\cos 9t + 2\sin 5t\sin 9t)}$$

$$|x(t)| = \sqrt{2 + 2\cos 4t}$$

$$|x(t)| = \sqrt{2(1 + \cos 4t)}$$

$$|x(t)| = \sqrt{2(1 + 2\cos^2 2t - 1)}$$

$$|x(t)| = \sqrt{2 \times 2\cos^2 2t}$$

$$|x(t)| = |2\cos 2t|$$

The fundamental period of $\cos(2t)$ is,

$$T = \frac{2\pi}{2} = \pi$$

But for $|\cos(2t)|$, the period is halved.

$$T = \frac{\pi}{2} \text{ sec}$$

$$\boxed{\text{(B) } \frac{\pi}{2}}$$

Q1.23.1 NAT 2023 1M Ans: -3

Given:

$$x(n) = u(-n+5) - u(n+3)$$

The signal $x(n)$ can be plotted as shown below,

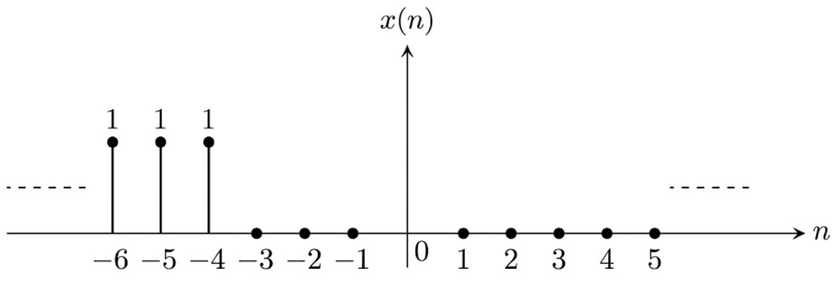

Fig. Solution Q1.23.1

So, the smallest value of n is -3 for $x(n)$ to be zero.
Hence, the correct answer is -3.

$$\boxed{-3.}$$

Q1.23.2 MCQ 2023 1M Ans: C

Theory: For an LTI system, the steady-state response to a sinusoidal input is determined by the magnitude and phase of the frequency response $H(j\omega)$.
The transfer function is

$$H(s) = \frac{s-\pi}{s+\pi}.$$

Since

$$y(t) = \sin(\pi t),$$

the excitation frequency is

$$\omega = \pi \text{ rad/s}.$$

Evaluating the frequency response,

$$H(j\omega) = \frac{j\omega - \pi}{j\omega + \pi}.$$

At $\omega = \pi$,

$$H(j\pi) = \frac{j\pi - \pi}{j\pi + \pi}.$$

The magnitude is

$$|H(j\pi)| = \frac{\sqrt{\pi^2 + \pi^2}}{\sqrt{\pi^2 + \pi^2}} = 1.$$

The phase is

$$\angle H(j\omega) = 180^\circ - 2\tan^{-1}\left(\frac{\omega}{\pi}\right).$$

Substituting $\omega = \pi$,

$$\angle H(j\pi) = 180^\circ - 2\tan^{-1}(1) = 180^\circ - 90^\circ = 90^\circ.$$

Since

$$Y(j\omega) = H(j\omega)X(j\omega),$$

the output leads the input by 90°. Therefore, the input must lag the output by 90°.
Therefore,

$$\boxed{(C)x(t) = \sin\left(\pi t - \frac{\pi}{2}\right)u(t)}$$

Key Points

- Evaluate the transfer function at the excitation frequency to obtain the steady-state gain and phase.
- The output phase equals the input phase plus the phase of $H(j\omega)$.
- Since $|H(j\pi)| = 1$, only the phase shift affects the input signal.

Mistakes to be avoided

- Do not substitute $s = \pi$; use $s = j\omega$.
- Do not confuse whether the input leads or lags the output; use $Y = HX$.
- Verify both magnitude and phase before selecting the input waveform.

Q1.23.3 NAT 2023 1M Ans: 25

Theory: Time scaling changes the period of a continuous-time periodic signal according to $T' = \frac{T}{|a|}$ for $x(at)$.

Given,

$$y(t) = x(4t),$$

and the fundamental period of $x(t)$ is

$$T_x = 100 \text{ s}.$$

If $x(t)$ has fundamental period T, then

$$x(at)$$

has fundamental period

$$T' = \frac{T}{|a|}.$$

Hence,

$$T_y = \frac{100}{4} = 25 \text{ s}.$$

Therefore,

$$\boxed{25}$$

Key Points

- Time compression by a factor of a reduces the period by the same factor.
- For $x(at)$, the new period is $T/|a|$.
- Always use the magnitude $|a|$ while computing the new period.

Mistakes to be avoided

- Do not multiply the period by the scaling factor; divide by $|a|$.
- Do not ignore the absolute value of the time-scaling factor.
- Ensure that the given period is the fundamental period before applying the formula.

Q1.23.4 MCQ 2023 2M Ans: A ● ● ○

Theory: For a sinusoidal input, the steady-state output of an LTI system is obtained by evaluating the frequency response $H(j\omega)$ at the input frequency.
Given,

$$h(t) = \delta(t) + 0.5\delta(t-4),$$

and

$$x(t) = \cos\left(\frac{7\pi}{4}t\right).$$

The frequency response is

$$H(j\omega) = 1 + 0.5e^{-j4\omega}.$$

The input frequency is

$$\omega = \frac{7\pi}{4}.$$

Hence,

$$H\left(j\frac{7\pi}{4}\right) = 1 + 0.5e^{-j7\pi} = 1 + 0.5(-1) = 0.5.$$

Since the frequency response is purely real,

$$\angle H\left(j\frac{7\pi}{4}\right) = 0^\circ,$$

there is no phase shift.
Therefore,

$$y(t) = H\left(j\frac{7\pi}{4}\right)x(t) = 0.5\cos\left(\frac{7\pi}{4}t\right).$$

Hence,

$$\boxed{(A) y(t) = 0.5\cos\left(\frac{7\pi}{4}t\right)}$$

Key Points

- The frequency response is the Fourier transform of the impulse response.
- A delayed impulse contributes a factor of $e^{-j\omega\tau}$ in the frequency response.
- For sinusoidal inputs, evaluate $H(j\omega)$ at the input frequency.

Mistakes to be avoided

- Do not substitute $e^{-j7\pi} = 1$; since 7 is odd, $e^{-j7\pi} = -1$.
- Do not perform time-domain convolution when the frequency-response method is simpler.
- Check whether $H(j\omega)$ introduces any phase shift before writing the output.

Q1.21.1 MCQ 2021 1M Ans: C ● ○ ○

Given:

$$x(t) = \sin\sqrt{2\pi t}$$

For a non-DC signal to be periodic, there must exist a constant T, such that,

$$x(t \pm T) = x(t) \qquad \cdots (i)$$

Equation (i) can be satisfied only if the power of t in $x(t)$ is 1.
As here the power of t in $x(t)$ is

$$\frac{1}{2}.$$

So, the signal is not periodic.
Hence, the correct option is (C).

$\boxed{\text{(C) Not periodic}}$

Q1.20.1 MCQ 2020 1M Ans: B

Given:

$$x[n] = \sin(2\pi n)u[n],$$

where $u[n] = 1$ for $n \geq 0$.

$$x[0] = \sin 2\pi(0)\, u[0] = 0$$

$$x[1] = \sin 2\pi(1)\, u[1] = 0$$

$$x[2] = \sin 2\pi(2)\, u[2] = 0$$

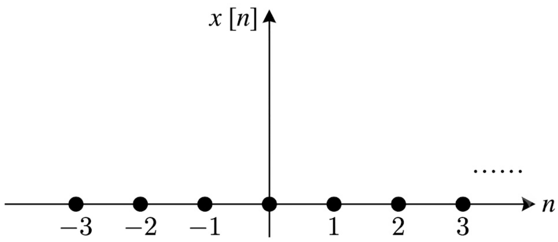

Fig. Solution Q1.20.1

$x[n]$ is a constant (DC) signal of zero value where it repeats its value zero with each value of n. So, the time period = 1.
Hence, the correct option is (B).

$\boxed{\text{(B) 1}}$

Q1.20.2 NAT 2020 1M Ans: 0

Theory: Convolution with the unit impulse sequence leaves the original sequence unchanged.
Given,

$$g[n] = \delta[n].$$

Hence,

$$y[n] = h[n] \otimes g[n] = h[n] \otimes \delta[n] = h[n].$$

The sequence $h[n]$ is

$$h[n] = \begin{cases} 1, & n = 0, 3, 6, 9, \ldots \\ 0, & \text{otherwise.} \end{cases}$$

Since 2 is not a multiple of 3,

$$h[2] = 0.$$

Therefore,

$$y[2] = h[2] = 0.$$

$\boxed{\text{y[2]=0}}$

Key Points

- The sequence $g[n]$ is the unit impulse sequence $\delta[n]$.
- Convolution with $\delta[n]$ satisfies $x[n] \otimes \delta[n] = x[n]$.
- The sequence $h[n]$ is nonzero only at indices $0, 3, 6, \ldots$.

Mistakes to be avoided

- Do not perform lengthy convolution when one sequence is $\delta[n]$.
- Do not confuse convolution with pointwise multiplication.
- Check whether the required index belongs to the set of nonzero indices before evaluating the sequence.

Q1.19.1 NAT 2019 2M Ans: 6

Given:

$$x[n] = e^{j\left(\frac{5\pi}{3}\right)n} + e^{j\left(\frac{\pi}{4}\right)n}$$

For down-sampling by 4,

$$n = 4n$$

$$x[4n] = e^{j\left(\frac{5\pi}{3}\right)4n} + e^{j\left(\frac{\pi}{4}\right)4n}$$

$$\omega_1 = \frac{20\pi}{3}, \qquad \omega_2 = \pi$$

$$N_1 = \frac{2\pi k}{20\pi/3} = \frac{3k}{10}$$

Putting $k = 10$,

$$N_1 = 3$$

$$N_2 = \frac{2\pi k}{\pi} = 2k$$

Putting $k = 1$,

$$N_2 = 2$$

$$(N_1, N_2) = (3, 2)$$

$$\text{Fundamental period} = \text{LCM}(N_1, N_2)$$

$$\text{Fundamental period} = 6$$

Hence, the fundamental period of the down-sampled signal $x_d[n]$ is 6.

$$\boxed{6}$$

Q1.18.1 MCQ 2018 1M Ans: A

Given:

$$x(t) = \begin{cases} 1, & |t| \le 2 \\ 0, & |t| > 2 \end{cases}$$

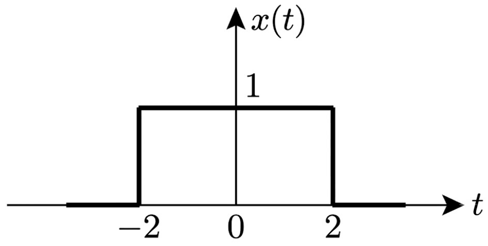

Fig. Solution Q1.18.1

$$I = \int_0^5 2x(t-3)\delta(t-4)\,dt$$

$$I = 2x(4-3)\int_0^5 \delta(t-4)\,dt$$

Since, area under the impulse function is 1,

$$I = 2x(1) \times 1 = 2$$

Hence, the correct option is (A).

$$\boxed{\text{(A) } 2}$$

Q1.18.2 MCQ 2018 1M Ans: D

Given:

$$z(t) = x(-t) + y(2t+1)$$

where, $x(t)$ and $y(t)$ are periodic signals with period 3 secs.
Period of $x(-t)$ is 3 sec.

$$\text{Period of } y\left[2\left(t + \frac{1}{2}\right)\right] = \frac{3}{2} = 1.5$$

Now, period of $z(t)$ is given by,

$$T = \text{LCM}(3, 1.5) = 3$$

Hence, the correct option is (D).

$$\boxed{\text{(D) } 3}$$

Key Points

- Time inversion and time shifting do not change the time period of a signal.
- If signal $x(t)$ has period T_0, after time scaling the signal $x(t)$ by factor k, i.e., $x(kt)$,

$$\text{Time period} = \frac{T_0}{k}.$$

Q1.17.1 NAT 2017 1M Ans: 1

Given:

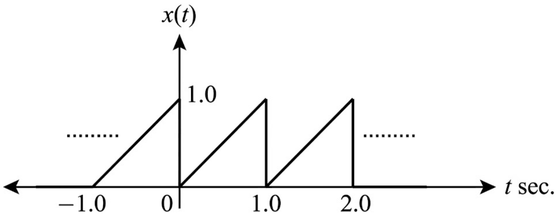

Fig. Solution Q1.17.1

Fundamental period,

$$T_0 = 1 \text{ sec}$$

Fundamental frequency,

$$f_0 = \frac{1}{T_0} = 1 \text{ Hz}$$

Hence, fundamental frequency of signal $x(t)$ is 1 Hz.

$$\boxed{1}$$

Q1.16.1 NAT 2016 2M Ans: 8

Given:

$$x(n) = \sin\left(\frac{301}{4}\pi n\right)$$

where,

$$\omega_0 = \frac{301}{4}\pi \text{ rad/sec}$$

Fundamental time period of discrete-time signal is given by,

$$N_0 = m \times \frac{2\pi}{\omega_0}$$

where, m is the minimum integer number to de-rationalize time period N_0.

$$N_0 = m \times \frac{2\pi}{301\pi/4}$$

$$N_0 = m \times \frac{8}{301}$$

For $m = 301$,

$$N_0 = 8$$

Hence, the fundamental period N_0 is 8.

$$\boxed{8}$$

Q1.15.1 NAT 2015 2M Ans: 6 ● ○ ○

Given:

$$x(t) = 2\cos\left(\frac{2\pi}{3}t\right) + \cos(\pi t)$$

Method Method 1

$$\omega_1 = \frac{2\pi}{3} \text{ rad/sec}, \qquad T_1 = \frac{2\pi}{\omega_1} = \frac{2\pi}{2\pi/3} = 3 \text{ sec}$$

$$\omega_2 = \pi \text{ rad/sec}, \qquad T_2 = \frac{2\pi}{\omega_2} = \frac{2\pi}{\pi} = 2 \text{ sec}$$

Fundamental time period of T_1 and T_2 is calculated as,

	×1	×2	×3
$T_1 = 3$	3	6	9
$T_2 = 2$	2	4	6

Fundamental time period,

$$T_0 = 6 \text{ sec}$$

Hence, fundamental time period of the signal is 6 sec.

Method Method 2

$$\omega_1 = \frac{2\pi}{3} \text{ rad/sec}$$

$$\omega_2 = \pi \text{ rad/sec}$$

Fundamental angular frequency is given by,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2)$$

$$\omega_0 = \text{HCF}\left(\frac{2\pi}{3}, \pi\right) = \frac{\pi}{3}$$

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{\pi/3} = 6$$

Hence, fundamental time period of the signal is 6 sec.

$$\boxed{6}$$

Q1.13.1 MCQ 2013 1M Ans: A ● ○ ○

Given:

$$v(t) = 30\sin 100t + 10\cos 300t + 6\sin\left(500t + \frac{\pi}{4}\right)$$

$$\omega_1 = 100 \text{ rad/sec}$$

$$\omega_2 = 300 \text{ rad/sec}$$

$$\omega_3 = 500 \text{ rad/sec}$$

Fundamental angular frequency is given by,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2, \omega_3)$$

$$\omega_0 = \text{HCF}(100, 300, 500)$$

$$\omega_0 = 100 \text{ rad/sec}$$

Hence, the correct option is (A).

$$\boxed{\text{(A) } 100}$$

Q1.11.1 MCQ 2011 1M Ans: C ● ○ ○

Given:

$$x(n) = \sin \omega_0 n$$

For a discrete-time signal to be periodic, the necessary condition is given by,

$$\frac{m}{N_0} = \frac{\omega_0}{2\pi} = \frac{p}{q} = \text{Ratio of integers.}$$

From the given options, only option (C) satisfies the above condition.
Hence, the correct option is (C).

$$\boxed{\text{(C) } \frac{2\pi}{\omega_0} \text{ to be a ratio of integers}}$$

Q1.11.2 MCQ 2011 1M Ans: A ● ○ ○

Given:

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2} \delta(1-2t)\, dt$$

Method Method 1

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2} \times \frac{1}{2}\delta\left(t - \frac{1}{2}\right) dt$$

$$I = \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{1}{2}\right)^2 e^{-(1/2)^2/2} \delta\left(t - \frac{1}{2}\right) dt$$

$$I = \frac{1}{8\sqrt{2\pi}} e^{-1/8} \int_{-\infty}^{\infty} \delta\left(t - \frac{1}{2}\right) dt$$

Since, area under the impulse function is 1,

$$I = \frac{1}{8\sqrt{2\pi}} e^{-1/8}$$

Hence the correct option is (A).

Method Method 2

Using integration property,

$$\int_{-\infty}^{\infty} x(t)\delta(t-t_0)\,dt = x(t_0)$$

$$I = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} t^2 e^{-t^2/2}\delta(1-2t)\,dt$$

$$I = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} t^2 e^{-t^2/2}\times\frac{1}{2}\delta\left(t-\frac{1}{2}\right)dt$$

$$I = \frac{1}{2\sqrt{2\pi}}\left(\frac{1}{2}\right)^2 e^{-(1/2)^2/2}$$

$$I = \frac{1}{8\sqrt{2\pi}}e^{-1/8}$$

Hence the correct option is (A).

$$\boxed{\text{(A) } \frac{1}{8\sqrt{2\pi}}e^{-1/8}}$$

Q1.10.1 MCQ 2010 1M Ans: B ● ○ ○

Given:

$$I = \int_{-\infty}^{\infty}\delta\left(t-\frac{\pi}{6}\right)6\sin t\,dt$$

Method Method 1

$$I = \int_{-\infty}^{\infty} 6\sin\left(\frac{\pi}{6}\right)\delta\left(t-\frac{\pi}{6}\right)dt$$

$$I = 6\times\frac{1}{2}\int_{-\infty}^{\infty}\delta\left(t-\frac{\pi}{6}\right)dt$$

Since, area under the impulse function is 1,

$$I = 3\times 1 = 3$$

Hence the correct option is (B).

Method Method 2

Using integration property,

$$\int_{-\infty}^{\infty} x(t)\delta(t-t_0)\,dt = x(t_0)$$

$$I = \int_{-\infty}^{\infty}\delta\left(t-\frac{\pi}{6}\right)6\sin t\,dt$$

$$I = 6\sin t\Big|_{t=\pi/6} = 3$$

Hence, the correct option is (B).

$$\boxed{\text{(B) } 3}$$

Q1.09.1 MCQ | 2009 | 1M | Ans: C | ● ○ ○

Given:

$$x(t) = 2\sin 2\pi t + 3\sin 3\pi t$$

where,

$$\omega_1 = 2\pi \text{ rad/sec}$$

and

$$\omega_2 = 3\pi \text{ rad/sec}$$

Fundamental angular frequency,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2) = \text{HCF}(2\pi, 3\pi)$$

$$\omega_0 = \pi \text{ rad/sec}$$

Fundamental time period,

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{\pi} = 2 \text{ sec}$$

Hence, the correct option is (C).

$$\boxed{\text{(C) 2 sec}}$$

Q1.08.1 MCQ | 2008 | 1M | Ans: D | ● ○ ○

Given:

$$x(n) = e^{j\left(\frac{5\pi}{6}\right)n}$$

where,

$$\omega_0 = \frac{5\pi}{6} \text{ rad/sec}$$

For a discrete-time signal to be periodic,

$$\frac{m}{N_0} = \frac{\omega_0}{2\pi} = \text{Ratio of integers}$$

$$N_0 = \frac{2\pi}{\omega_0} \times m = \frac{2\pi}{5\pi/6} \times m$$

$$N_0 = \frac{12}{5} \times m$$

where, m is the minimum integer number to de-rationalize the time period N_0.
For $m = 5$,

$$N_0 = 12$$

Hence, the correct option is (D).

$$\boxed{\text{(D) 12}}$$

Q1.07.1 MCQ 2007 2M Ans: C ● ○ ○

Given:

$$x(n) = \left(\frac{1}{3}\right)^n u(n)$$

where,

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

and

$$y(n) = x(-n), \qquad -\infty < n < \infty$$

$$\sum_{n=-\infty}^{\infty} y(n) = \sum_{n=-\infty}^{\infty} x(-n)$$

$$\sum_{n=-\infty}^{\infty} y(n) = \sum_{n=-\infty}^{\infty} \left(\frac{1}{3}\right)^{-n} u(-n)$$

$$\sum_{n=-\infty}^{\infty} y(n) = \sum_{n=-\infty}^{0} \left(\frac{1}{3}\right)^{-n}$$

$$\sum_{n=-\infty}^{\infty} y(n) = 1 + \left(\frac{1}{3}\right)^1 + \left(\frac{1}{3}\right)^2 + \cdots$$

$$\sum_{n=-\infty}^{\infty} y(n) = \frac{1}{1 - \frac{1}{3}} = \frac{3}{2}$$

Hence, the correct option is (C).

$$\boxed{(C)\ \frac{3}{2}}$$

Q1.07.2 MCQ 2007 1M Ans: A ● ○ ○

Given:

$$x(t) = \left(1 + 0.5\cos 40\pi t\right)\cos 200\pi t$$

$$x(t) = \cos 200\pi t + \frac{0.5}{2}\left[\cos 240\pi t + \cos 160\pi t\right]$$

$$x(t) = \cos 200\pi t + 0.25\cos 240\pi t + 0.25\cos 160\pi t$$

$$x(t) = 0.25\cos 160\pi t + \cos 200\pi t + 0.25\cos 240\pi t$$

where,

$$\omega_1 = 160\pi \text{ rad/sec}$$

$$\omega_2 = 200\pi \text{ rad/sec}$$

$$\omega_3 = 240\pi \text{ rad/sec}$$

Fundamental angular frequency,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2, \omega_3)$$

$$\omega_0 = \text{HCF}(160\pi, 200\pi, 240\pi)$$

$$\omega_0 = 40\pi \text{ rad/sec}$$

Fundamental frequency,

$$f_0 = \frac{\omega_0}{2\pi} = \frac{40\pi}{2\pi} = 20 \text{ Hz}$$

Hence, the correct option is (A).

$$\boxed{\text{(A) } 20}$$

Q1.06.1 MCQ | 2006 | 2M | Ans: A | ● ○ ○

Given:

$$x(t) = \cos(1.2\pi t) + \cos(2\pi t) + \cos(2.8\pi t)$$

$$\omega_1 = 1.2\pi \text{ rad/sec} = \frac{6\pi}{5} \text{ rad/sec}$$

$$\omega_2 = 2\pi \text{ rad/sec}$$

$$\omega_3 = 2.8\pi \text{ rad/sec} = \frac{14\pi}{5} \text{ rad/sec}$$

Fundamental angular frequency is given by,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2, \omega_3)$$

$$\omega_0 = \text{HCF}\left(\frac{6\pi}{5}, \frac{2\pi}{1}, \frac{14\pi}{5}\right)$$

$$\omega_0 = \frac{\text{HCF}[6\pi, 2\pi, 14\pi]}{\text{LCM}[5, 1, 5]}$$

$$\omega_0 = \frac{2\pi}{5} \text{ rad/sec}$$

$$f_0 = \frac{\omega_0}{2\pi} = \frac{\frac{2\pi}{5}}{2\pi} = \frac{1}{5} = 0.2 \text{ Hz}$$

Hence, the correct option is (A).

$$\boxed{\text{(A) } 0.2 \text{ Hz}}$$

Q1.06.2 MCQ | 2006 | 1M | Ans: C | ● ○ ○

Given: 2 V DC voltage and 4 V peak-to-peak square wave signal are superimposed i.e.

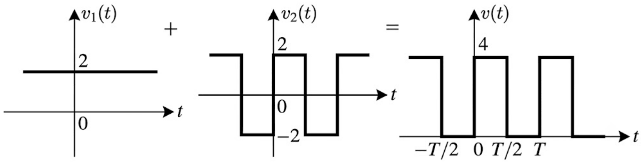

Fig. Solution Q1.06.2

RMS value of superimposed signal is given by,

$$v_{\text{rms}} = \sqrt{\frac{1}{T}\int_{-T/2}^{T/2} v^2(t)\,dt}$$

$$v_{\text{rms}} = \sqrt{\frac{1}{T}\left[\int_{-T/2}^{0} 0\,dt + \int_{0}^{T/2} 4^2\,dt\right]}$$

$$v_{\text{rms}} = \sqrt{\frac{1}{T}\left[16 \times \frac{T}{2}\right]} = \sqrt{8}\text{ V}$$

Hence, the correct option is (C).

$$\boxed{(C)\ \sqrt{8}\text{ V}}$$

Q1.05.1 MCQ | 2005 | 1M | Ans: A | ● ○ ○

Given:

$$x(n) = 3\sin(1.3\pi n + 0.5\pi) + 5\sin(1.2\pi n)$$

$$\omega_1 = 1.3\pi \text{ rad/sec}, \qquad \omega_2 = 1.2\pi \text{ rad/sec}$$

Fundamental angular frequency is given by,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2)$$

$$\omega_0 = \text{HCF}(1.3\pi, 1.2\pi)$$

$$\omega_0 = \text{HCF}\left(\frac{13\pi}{10}, \frac{12\pi}{10}\right)$$

$$\omega_0 = \frac{\text{HCF}[12\pi, 13\pi]}{\text{LCM}[10, 10]}$$

$$\omega_0 = \frac{\pi}{10}\text{ rad/sec}$$

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{\pi/10} = 20$$

Hence, the correct option is (A).

$$\boxed{\text{(A) } 20}$$

Key Points

- Fundamental angular frequency:
$$\omega_0 = \text{HCF}(\omega_1, \omega_2, \omega_3, \ldots)$$
- Fundamental period:
$$T = \frac{2\pi}{\omega_0}$$

Q1.01.1

MCQ | 2001 | 2M | Ans: D

Power of a discrete-time signal is

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x(n)|^2$$

Option (A): $x(n) = u(n)$

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=0}^{N} 1 = \lim_{N\to\infty} \frac{N+1}{2N+1} = \frac{1}{2}$$

Hence, unit step sequence is a power signal.
Option (B): $x(n) = e^{j\omega_0 n}$

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |e^{j\omega_0 n}|^2$$

$$= \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} 1 = 1$$

Hence, $e^{j\omega_0 n}$ is a power signal.
Option (C): A periodic sequence.
Every periodic sequence has finite average power. Hence, it is a power signal.
Option (D): $x(n) = nu(n)$

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=0}^{N} n^2$$

$$= \lim_{N\to\infty} \frac{N(N+1)(2N+1)}{6(2N+1)}$$

$$= \lim_{N\to\infty} \frac{N(N+1)}{6} = \infty$$

Thus, the unit ramp sequence is *not* a power signal.

$$\boxed{\text{(D) Unit ramp sequence}}$$

SOLUTIONS

II

Basics of Systems

Topics

1. Continuous-Time Convolution
2. Discrete-Time Convolution
3. Linear and Non-Linear Systems
4. Time Variant and Time-Invariant Systems
5. LTI Systems - Definition and Properties
6. Causal and Non-Causal Systems
7. Static and Dynamic Systems
8. Stable and Unstable Systems
9. Invertible and Non-Invertible Systems
10. Systems Involving Differential Equations

Q2.26.1 MCQ 2026 2M Ans: A

(M) $y(t) = x^2(t)$
Linearity:
Let $x(t) = ax_1(t) + bx_2(t)$.

$$y(t) = [ax_1(t) + bx_2(t)]^2$$

$$= a^2x_1^2(t) + b^2x_2^2(t) + 2ab\,x_1(t)x_2(t)$$

Since

$$[ax_1(t) + bx_2(t)]^2 \neq ax_1^2(t) + bx_2^2(t),$$

the system is non-linear.
Time Invariance:
For $x_1(t) = x(t - t_0)$,

$$y_1(t) = [x(t - t_0)]^2$$

and

$$y(t - t_0) = [x(t - t_0)]^2.$$

Hence $y_1(t) = y(t - t_0)$, so the system is time-invariant.

$$(M) \to (IV)$$

(N) $y(t) = x(-t)$
Linearity:

$$y(t) = ax_1(-t) + bx_2(-t) = ay_1(t) + by_2(t)$$

Hence, the system is linear.
Time Invariance:
For $x_1(t) = x(t - t_0)$,

$$y_1(t) = x(-t - t_0)$$

while

$$y(t - t_0) = x(-t + t_0).$$

Since

$$x(-t - t_0) \neq x(-t + t_0),$$

the system is time-varying.

$$(N) \to (III)$$

(O) $y(t) = |x(3 - t)|$
Linearity:

$$y(t) = |ax_1(3 - t) + bx_2(3 - t)|$$

but

$$|ax_1(3-t) + bx_2(3-t)| \neq a|x_1(3-t)| + b|x_2(3-t)|.$$

Hence, the system is non-linear.
Time Invariance:
For $x_1(t) = x(t - t_0)$,

$$y_1(t) = |x(3 - t - t_0)|$$

while

$$y(t - t_0) = |x(3 - t + t_0)|.$$

Since

$$|x(3 - t - t_0)| \neq |x(3 - t + t_0)|,$$

the system is time-varying.

$$(O) \rightarrow (II)$$

(P) $y(t) = x(t-3)$
Linearity:

$$y(t) = ax_1(t-3) + bx_2(t-3) = ay_1(t) + by_2(t)$$

Hence, the system is linear.
Time Invariance:
For $x_1(t) = x(t - t_0)$,

$$y_1(t) = x(t - t_0 - 3)$$

and

$$y(t - t_0) = x((t - t_0) - 3) = x(t - t_0 - 3).$$

Hence $y_1(t) = y(t - t_0)$, so the system is time-invariant.

$$(P) \rightarrow (I)$$

Therefore,

$$M-IV, \quad N-III, \quad O-II, \quad P-I$$

$$\boxed{\text{(A) M-IV, N-III, O-II, P-I}}$$

Q2.25.1 MSQ | 2025 | 2M | Ans: A,C | ● ○ ○

For an LTI system, an eigenfunction satisfies

$$T\{x(t)\} = \lambda x(t)$$

where λ is a scalar.
The eigenfunctions of continuous-time LTI systems are complex exponentials of the form

$$x(t) = e^{st}.$$

Option (A):

$$x(t) = e^{j\frac{2\pi}{3}t}$$

This is a complex exponential. Hence, it is an eigenfunction.
Option (B):

$$x(t) = \cos\left(\frac{2\pi}{3}t\right)$$

A cosine is a sum of two complex exponentials and is not an eigenfunction by itself.
Option (C):

$$x(t) = 2^t = e^{(\ln 2)t}$$

This is an exponential function of the form e^{st}. Hence, it is an eigenfunction.
Option (D):

$$x(t) = \sin\left(\frac{2\pi}{3}t\right)$$

A sine is also a combination of complex exponentials and is not an eigenfunction by itself.

$$\boxed{\text{(A), (C)}}$$

Q2.25.2 MCQ | 2025 | 2M | Ans: C

Given,

$$y[n] = \frac{1}{\alpha}y[n-1] + x[n], \qquad \alpha > 1$$

and

$$x[n] = \delta[n-p], \qquad p > 10.$$

Substituting $x[n]$,

$$y[n] = \frac{1}{\alpha}y[n-1] + \delta[n-p]$$

For $n = p$,

$$y[p] = \frac{1}{\alpha}y[p-1] + \delta[0]$$

$$y[p] = \frac{1}{\alpha}y[p-1] + 1$$

For $n = p+1$,

$$y[p+1] = \frac{1}{\alpha}y[p] + \delta[1]$$

$$y[p+1] = \frac{1}{\alpha}y[p]$$

Substituting $y[p]$,

$$y[p+1] = \frac{1}{\alpha}\left(\frac{1}{\alpha}y[p-1] + 1\right)$$

$$y[p+1] = \frac{1}{\alpha^2}y[p-1] + \frac{1}{\alpha}$$

Since the system is initially at rest and $p > 10$,

$$y[p-1] = 0$$

Therefore,

$$y[p+1] = \frac{1}{\alpha}$$

$$\boxed{\text{(C) } \frac{1}{\alpha}}$$

Q2.22.1 MCQ 2022 1M Ans: D ● ○ ○

Given,

$$y(t) = x(t)\sin(2\pi t)$$

Linearity:
For inputs $x_1(t)$ and $x_2(t)$,

$$y_1(t) = x_1(t)\sin(2\pi t), \qquad y_2(t) = x_2(t)\sin(2\pi t)$$

Then,

$$ay_1(t) + by_2(t) = ax_1(t)\sin(2\pi t) + bx_2(t)\sin(2\pi t)$$

$$= \left[ax_1(t) + bx_2(t)\right]\sin(2\pi t)$$

For input

$$x_3(t) = ax_1(t) + bx_2(t),$$

the output is

$$y_3(t) = \left[ax_1(t) + bx_2(t)\right]\sin(2\pi t).$$

Hence,

$$y_3(t) = ay_1(t) + by_2(t)$$

and the system is linear.
Time Invariance:
For delayed input $x(t-t_0)$,

$$y_1(t) = x(t-t_0)\sin(2\pi t)$$

The delayed version of the original output is

$$y(t-t_0) = x(t-t_0)\sin\big(2\pi(t-t_0)\big)$$

Since

$$x(t-t_0)\sin(2\pi t) \neq x(t-t_0)\sin\big(2\pi(t-t_0)\big),$$

we have

$$y_1(t) \neq y(t-t_0).$$

Therefore, the system is time-varying.

(D) Linear and time-variant

Q2.19.1 MCQ 2019 1M Ans: A ● ○ ○

Given,

$$y[n] = \alpha y[n-1] + x[n]$$

Taking the z-transform on both sides,

$$Y(z) = \alpha z^{-1} Y(z) + X(z)$$

$$Y(z)\left(1 - \alpha z^{-1}\right) = X(z)$$

Hence, the transfer function is

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - \alpha z^{-1}}$$

The pole is located at

$$z = \alpha.$$

For a causal discrete-time system to be BIBO stable, all poles must lie inside the unit circle.
Therefore,

$$|\alpha| < 1.$$

(A) $|\alpha| < 1$

Q2.14.1 MCQ 2014 1M Ans: A ● ○ ○

Given:

$$h(n) = \begin{cases} \dfrac{\omega_c}{\pi}, & n = 0 \\ \dfrac{\sin(\omega_c n)}{\pi n}, & n \neq 0 \end{cases}$$

Condition for causal system:
Impulse response of the system should satisfy

$$h(n) = 0, \qquad n < 0$$

Since,

$$h(n) = \frac{\sin(\omega_c n)}{\pi n}, \qquad n \neq 0$$

the impulse response is non-zero for $n < 0$.
Therefore, the given system is a non-causal system.

$$h(n) = \frac{\sin(\omega_c n)}{\pi n}$$

Discrete-time Fourier transform of $h(n)$ is shown below.

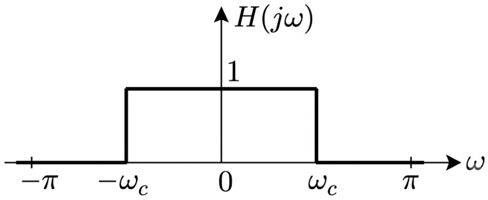

Fig. Solution Q2.14.1

Above figure represents an ideal low-pass filter.

$$\boxed{\text{(A) Non-causal, low-pass filter}}$$

Q2.13.1 MCQ 2013 1M Ans: C

For z-plane, condition for stability for a causal system is that all the poles must lie inside

$$|z| = 1.$$

For s-plane, condition for stability for a causal system is that all the poles must lie in the negative half of s-plane.

Statement 1: All the poles of the system must lie on the left side of the $j\omega$ axis.
From above discussion, we can clearly show that statement 1 is true.
Statement 2: Zeros of the system can lie anywhere in the s-plane.
Since stability of the system depends on the poles, hence zeros can lie anywhere in s-plane for a stable system.
Statement 3: All the poles must lie within $|s| = 1$.
There is no such condition for stability in s-plane.
Therefore, statement 3 is wrong.
Statement 4: All the roots of the characteristic equation must be located on the left side of the $j\omega$ axis.
From above discussion, statement 4 is true.
Hence, the correct option is (C).

$$\boxed{\text{(C) All the poles must lie within } |s| = 1.}$$

Q2.13.2 MCQ 2013 2M Ans: B

Given: Impulse response,

$$h(t) = \delta(t-1) + \delta(t-3)$$

$$\frac{d}{dt}(\text{Step response}) = \text{Impulse response}$$

$$\frac{d}{dt}[s(t)] = h(t)$$

Thus,

$$s(t) = \int_{-\infty}^{t} h(t)\,dt$$

$$s(t) = \int_{-\infty}^{t} \left[\delta(t-1) + \delta(t-3)\right] dt$$

$$s(t) = u(t-1) + u(t-3)$$

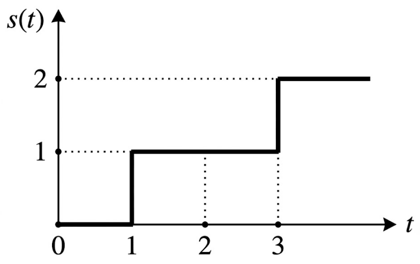

Fig. Solution Q2.13.2

$$s(t)\Big|_{t=2} = 1$$

Hence, the correct option is (B).

$$\boxed{\text{(B) } 1}$$

Q2.12.1 MCQ 2012 2M Ans: D ● ○ ○

Given,

$$y(t) = \int_{-\infty}^{t} x(\tau)\cos(3\tau)\,d\tau$$

(i) Time invariance:
Let the input be delayed by t_0. Then,

$$y(t, t_0) = \int_{-\infty}^{t} x(\tau - t_0)\cos(3\tau)\,d\tau$$

Let

$$\tau - t_0 = \tau_1$$

Then,

$$y(t, t_0) = \int_{-\infty - t_0}^{t - t_0} x(\tau_1)\cos\left[3(t_0 + \tau_1)\right]d\tau_1 \tag{i}$$

Replacing t by $t - t_0$,

$$y(t - t_0) = \int_{-\infty}^{t - t_0} x(\tau)\cos(3\tau)\,d\tau \tag{ii}$$

From (i) and (ii),

$$y(t, t_0) \neq y(t - t_0)$$

Hence, the system is time-varying.
(ii) Stability:
Let

$$x(t) = \cos(3t)$$

which is a bounded input.
Then,

$$y(t) = \int_{-\infty}^{t} \cos^2(3\tau)\,d\tau$$

Using

$$\cos^2(3\tau) = \frac{1+\cos(6\tau)}{2},$$

$$y(t) = \int_{-\infty}^{t} \frac{1+\cos(6\tau)}{2}\, d\tau$$

$$y(t) = \frac{1}{2}\left[\tau + \frac{\sin(6\tau)}{6}\right]_{-\infty}^{t} = \infty$$

Thus, for a bounded input, the output is unbounded.
Hence, the system is not stable.

$$\boxed{\text{(D) not time-invariant and not stable}}$$

Q2.11.1 MCQ | 2011 | 1M | Ans: D

Given,

$$y(t) = \frac{d}{dt}\left\{e^{-t}x(t)\right\}$$

(i) Linearity:
For input $x_1(t)$,

$$y_1(t) = \frac{d}{dt}\left\{e^{-t}x_1(t)\right\}$$

Multiplying by scalar a,

$$a\,y_1(t) = a\frac{d}{dt}\left\{e^{-t}x_1(t)\right\} \tag{i}$$

For input $x_2(t)$,

$$y_2(t) = \frac{d}{dt}\left\{e^{-t}x_2(t)\right\}$$

Multiplying by scalar b,

$$b\,y_2(t) = b\frac{d}{dt}\left\{e^{-t}x_2(t)\right\} \tag{ii}$$

Adding (i) and (ii),

$$a\,y_1(t) + b\,y_2(t) = a\frac{d}{dt}\left\{e^{-t}x_1(t)\right\} + b\frac{d}{dt}\left\{e^{-t}x_2(t)\right\} \tag{iii}$$

Now let the input be

$$a\,x_1(t) + b\,x_2(t)$$

with output $y_3(t)$.

$$y_3(t) = \frac{d}{dt}\left\{e^{-t}\left(a\,x_1(t) + b\,x_2(t)\right)\right\}$$

$$y_3(t) = a\frac{d}{dt}\left\{e^{-t}x_1(t)\right\} + b\frac{d}{dt}\left\{e^{-t}x_2(t)\right\} \tag{iv}$$

From (iii) and (iv),

$$y_3(t) = a\,y_1(t) + b\,y_2(t)$$

Therefore, the system satisfies superposition and is ***linear***.

(ii) Time Variance:

For delayed input $x(t - t_0)$,

$$y_1(t) = \frac{d}{dt}\left\{e^{-t}x(t - t_0)\right\} \tag{v}$$

The delayed version of the output is

$$y(t - t_0) = \frac{d}{dt}\left\{e^{-(t-t_0)}x(t - t_0)\right\} \tag{vi}$$

From (v) and (vi),

$$y_1(t) \neq y(t - t_0)$$

Hence, the system is ***time-varying***.

(iii) Stability:

For bounded input

$$x(t) = u(t),$$

$$y(t) = \frac{d}{dt}\left\{e^{-t}u(t)\right\}$$

$$y(t) = e^{-t}\delta(t) + u(t)(-e^{-t})$$

For bounded input, the output is unbounded due to the impulse term.
Therefore, the system is ***unstable***.

(iv) Memory:

A system is memoryless if the output depends only on the present input.
Differentiation depends on neighboring values of the input signal.
Hence, the system possesses ***memory*** and is a dynamic system.

$\boxed{\text{(D) The system has memory}}$

Q2.10.1 MCQ | 2010 | 2M | Ans: C | ● ○ ○

Given,

$$y(t) = \int_{-\infty}^{5t} x(\tau)\, d\tau$$

(i) Time Variance:

Let the input be delayed by t_0. Then the corresponding output is

$$y_1(t) = \int_{-\infty}^{5t} x(\tau - t_0)\, d\tau \tag{i}$$

Now delay the output $y(t)$ by t_0,

$$y(t - t_0) = \int_{-\infty}^{5(t-t_0)} x(\tau)\, d\tau \tag{ii}$$

From (i) and (ii),

$$y_1(t) \neq y(t - t_0)$$

Therefore, the system is time-varying.

(ii) Causality:

If the output of a system depends only on present and past inputs, then the system is causal. If it depends on future inputs also, then it is non-causal.

Given,

$$y(t) = \int_{-\infty}^{5t} x(\tau)\, d\tau$$

For $t = -1$,

$$y(-1) = \int_{-\infty}^{-5} x(\tau)\, d\tau$$

For $t = 1$,

$$y(1) = \int_{-\infty}^{5} x(\tau)\, d\tau$$

Here, it is clear that the output depends upon both past and future values of the input.

Therefore, the system is non-causal.

$$\boxed{\text{(C) time variant and non-causal}}$$

Q2.09.1 MCQ | 2009 | 2M | Ans: D | ● ○ ○

Given,

$$y(t) = \sum_{k=-\infty}^{\infty} x(kT)\, \delta(t - kT)$$

(i) Linearity:

For input $x_1(t)$, output will be $y_1(t)$.

$$y_1(t) = \sum_{k=-\infty}^{\infty} x_1(kT)\, \delta(t - kT)$$

Multiplying both sides by scalar quantity a,

$$a\, y_1(t) = a \sum_{k=-\infty}^{\infty} x_1(kT)\, \delta(t - kT) \tag{i}$$

For input $x_2(t)$, output will be $y_2(t)$.

$$y_2(t) = \sum_{k=-\infty}^{\infty} x_2(kT)\, \delta(t - kT)$$

Multiplying both sides by scalar quantity b,

$$b\, y_2(t) = b \sum_{k=-\infty}^{\infty} x_2(kT)\, \delta(t - kT) \tag{ii}$$

Adding equations (i) and (ii),

$$a\, y_1(t) + b\, y_2(t) = \sum_{k=-\infty}^{\infty} [a\, x_1(kT) + b\, x_2(kT)]\, \delta(t - kT) \tag{iii}$$

Now, let the system have input

$$a\, x_1(t) + b\, x_2(t)$$

and corresponding output be $y_3(t)$.

$$y_3(t) = \sum_{k=-\infty}^{\infty} [a\,x_1(kT) + b\,x_2(kT)]\,\delta(t-kT) \tag{iv}$$

From equations (iii) and (iv),

$$y_3(t) = a\,y_1(t) + b\,y_2(t)$$

Therefore, the system follows the principle of superposition. Hence, the system is linear.
(ii) Time Variance:
In a time invariant system, any delay provided in input must be reflected in output.
Let the input be delayed by t_0, then corresponding output is $y_1(t)$.

$$y_1(t) = \sum_{k=-\infty}^{\infty} x(kT-t_0)\,\delta(t-kT) \tag{v}$$

Now, output $y(t)$ is delayed by t_0,

$$y(t-t_0) = \sum_{k=-\infty}^{\infty} x(kT)\,\delta(t-t_0-kT) \tag{vi}$$

From equations (v) and (vi),

$$y_1(t) \neq y(t-t_0)$$

Thus, the system is time varying.

$$\boxed{\text{(D) linear and time-varying}}$$

Q2.08.1 MCQ 2008 1M Ans: D ● ○ ○

Time invariance:
In a time invariant system, any delay provided in input must be reflected in output.
(i) From option (A): $y(n) = n\,x(n)$
Let input be delayed by n_0. Then corresponding output is

$$y_1(n) = n\,x(n-n_0) \tag{i}$$

Now, output $y(n)$ is delayed by n_0,

$$y(n-n_0) = (n-n_0)x(n-n_0) \tag{ii}$$

From equations (i) and (ii),

$$y_1(n) \neq y(n-n_0)$$

Thus, the system is time varying.
(ii) From option (B): $y(n) = x(3n)$
Let input be delayed by n_0. Then corresponding output is

$$y_1(n) = x(3n-n_0) \tag{iii}$$

Now, output $y(n)$ is delayed by n_0,

$$y(n-n_0) = x\big[3(n-n_0)\big] \tag{iv}$$

From equations (iii) and (iv),

$$y_1(n) \neq y(n-n_0)$$

Thus, the system is time varying.
(iii) From option (C): $y(n) = x(-n)$
Let input be delayed by n_0. Then corresponding output is

$$y_1(n) = x(-n-n_0) \tag{v}$$

Now, output $y(n)$ is delayed by n_0,

$$y(n-n_0) = x\big[-(n-n_0)\big] \tag{vi}$$

From equations (v) and (vi),

$$y_1(n) \neq y(n-n_0)$$

Thus, the system is time varying.
(iv) From option (D): $y(n) = x(n-3)$
Let input be delayed by n_0. Then corresponding output is

$$y_1(n) = x(n-n_0-3) \tag{vii}$$

Now, output $y(n)$ is delayed by n_0,

$$y(n-n_0) = x\big[(n-3)-n_0\big] \tag{viii}$$

From equations (vii) and (viii),

$$y_1(n) = y(n-n_0)$$

Thus, the system is time invariant.

$$\boxed{\text{(D) } y[n] = x[n-3]}$$

Q2.01.1 MCQ | 2001 | 1M | Ans: D | ● ○ ○

Condition for causality of a LTI system:

$$h(n) = 0; \quad n < 0$$

(i) From option (A): $h(n) = a^n u(n-2)$
This signal will exist only for $n \geq 2$ i.e., for $n < 0$, $h(n) = 0$.
Therefore, it is causal.
(ii) From option (B): $h(n) = a^{n-2}u(n)$
This signal will exist only for $n \geq 0$ i.e., for $n < 0$, $h(n) = 0$.
Therefore, it is causal.
(iii) From option (C): $h(n) = a^{n+2}u(n)$
This signal will exist only for $n \geq 0$ i.e., for $n < 0$, $h(n) = 0$.
Therefore, it is causal.
(iv) From option (D): $h(n) = a^n u(n+2)$
This signal exists only for $n \geq -2$ i.e., for $n < 0$, $h(n) \neq 0$.
Therefore, it is non-causal.
Hence, the correct option is (D).

$$\boxed{\text{(D) } a^n u(n+2)}$$

SOLUTIONS

III

Z Transform

Topics

1. Introduction to Z-Transform
2. Z-Transform and ROC
3. Inverse Z-Transform
4. Important Properties of Z-Transform
5. Unilateral Z-Transform

Q6.24.1 NAT | 2024 | 2M | Ans: 0.75 | ○ ○ ○

Given $y[n] - 0.5\,y[n-2] = a\,x[n-4]$. Taking the Z-transform,

$$Y(z) - 0.5z^{-2}Y(z) = aZ^{-4}X(z) \;\Rightarrow\; H(Z) = \frac{Y(Z)}{X(Z)} = \frac{aZ^{-4}}{1-0.5Z^{-2}}.$$

Substituting $z = e^{j\omega}$ for the frequency response,

$$H(e^{j\omega}) = \frac{ae^{-j4\omega}}{1-0.5e^{-j2\omega}}.$$

At $\omega = \dfrac{\pi}{2}$, given $\left|H(e^{j\pi/2})\right| = 0.5$:

$$0.5 = \left|\frac{a\,e^{-j2\pi}}{1-0.5e^{-j\pi}}\right| = \frac{a}{1-0.5(-1)} = \frac{a}{1.5}.$$

$$a = 1.5 \times 0.5 = 0.75.$$

Hence, the value of a is **0.75**.

Q6.24.2 MCQ | 2024 | 2M | Ans: C | ● ● ○

Given

$$x(n) = \begin{cases} (0.2)^n, & 0 \le n \le 7 \\ 0, & \text{otherwise} \end{cases}$$

$$X(z) = \sum_{n=0}^{7}(0.2)^n z^{-n} = 1 + (0.2)z^{-1} + (0.2)^2z^{-2} + \cdots + (0.2)^7z^{-7}.$$

This is a finite sum of negative powers of z. It converges everywhere *except* $z = 0$, where the negative-power terms are undefined; there is no restriction at $z = \infty$ since the expression reduces to the finite constant term 1 there. Hence, the ROC is all values of z except $z = 0$, and the correct option is **(C)**.

Q6.23.1 MCQ | 2023 | 2M | Ans: B | ● ● ○

Given $F(z) = \dfrac{1}{1-z}$, expanded around $z = 2$. Let $t = z - 2$, so $z = 2 + t$.

$$F(t) = \frac{1}{1-(2+t)} = \frac{1}{-1-t} = \frac{-1}{1+t} = -\left(1 - t + t^2 - t^3 + \cdots\right).$$

$$F(z) = -1 + (z-2) - (z-2)^2 + (z-2)^3 - \cdots = \sum_{k=0}^{\infty}(-1)^{k+1}(z-2)^k.$$

Comparing with $F(z) = \sum_{k=0}^{\infty} a_k(z-2)^k$, we get $a_k = (-1)^{k+1}$.
Hence, the correct option is **(B)**.

Q6.18.1 MCQ | 2018 | 2M | Ans: D | ● ● ○

Given $Y(z) = 2 + 3z^{-1} + z^{-2}$ is the Z-transform of $y[n] = x[n] * h[n]$, and $p[n] = x[n] * h[n-2]$.
By the time-shifting property, convolving with a signal delayed by 2 samples multiplies the transform by z^{-2}:

$$P(z) = z^{-2}Y(z) = z^{-2}\left(2 + 3z^{-1} + z^{-2}\right) = 2z^{-2} + 3z^{-3} + z^{-4}.$$

Hence, the correct option is **(D)**.

Q6.17.1 MCQ 2017 1M Ans: C ● ○ ○

Given $x_1(n) = \{1, 1\}$ and $x_2(n) = \{1, 2\}$.

$$x_1(n) \xrightarrow{\text{Z.T.}} 1 + z^{-1}, \qquad x_2(n) \xrightarrow{\text{Z.T.}} 1 + 2z^{-1}.$$

By the convolution property,

$$x_1(n) \otimes x_2(n) = X_1(z)X_2(z) = (1 + z^{-1})(1 + 2z^{-1}) = 1 + 3z^{-1} + 2z^{-2}.$$

Hence, the correct option is **(C)**.

Q6.15.1 MCQ 2015 1M Ans: A ● ○ ○

Given $x(n) = \alpha^{|n|}$, $0 < |\alpha| < 1$. This can be split as

$$x(n) = \underbrace{\alpha^n u(n)}_{x_1(n)} + \underbrace{\alpha^{-n} u(-n-1)}_{x_2(n)}.$$

Using the standard pairs,

$$\alpha^n u(n) \overset{ZT}{\longleftrightarrow} \frac{1}{1 - \alpha z^{-1}}, \quad \text{ROC} : |z| > |\alpha|,$$

$$-\left(\frac{1}{\alpha}\right)^n u(-n-1) \overset{ZT}{\longleftrightarrow} \frac{1}{1 - \frac{1}{\alpha} z^{-1}}, \quad \text{ROC} : |z| < \frac{1}{|\alpha|}.$$

The common ROC exists (since $|\alpha| < 1$) and equals

$$|\alpha| < |z| < \frac{1}{|\alpha|}.$$

Hence, the correct option is **(A)**.

Q6.14.1 MCQ 2014 2M Ans: C ● ● ○

Given $\dfrac{b_0}{1 - z^{-1} + a_2 z^{-2}}$, a_2 real. Multiplying through by z^2, the poles satisfy

$$z^2 - z + a_2 = 0 \;\Rightarrow\; z = \frac{1 \pm \sqrt{1 - 4a_2}}{2}.$$

Checking each option for $|z| < 1$ (BIBO stability):

- $a_2 = 0.5$: $z = \dfrac{1 \pm j}{2}$, $|z| = \dfrac{1}{\sqrt{2}} \approx 0.707 < 1 \Rightarrow$ **stable**.
- $a_2 = 1.5$: $|z| = \dfrac{\sqrt{6}}{2} \approx 1.22 > 1 \Rightarrow$ unstable.
- $a_2 = -0.75$: $z = 1.5, -0.5$; $|1.5| > 1 \Rightarrow$ unstable.
- $a_2 = -1.5$: $z \approx 1.82, -0.82$; $|1.82| > 1 \Rightarrow$ unstable.

Hence, the correct option is **(C)**.

Q6.14.2 MCQ 2014 2M Ans: C

Given $H(z) = \dfrac{1-\frac{1}{3}z^{-1}}{1-\frac{1}{4}z^{-1}}$. $H(z)$ has a pole at $z = \frac{1}{4}$ and a zero at $z = \frac{1}{3}$.

For $H(z)$ itself to be causal and stable, its ROC must lie outside its outermost pole:

$$|z| > \frac{1}{4}.$$

Hence, the correct option is **(C)**.

Q6.13.1 MCQ 2013 1M Ans: D

Given $\dfrac{1-2z^{-1}}{1-0.5z^{-1}}$.

Zero: $1-2z^{-1} = 0 \Rightarrow z = 2$, which lies *outside* the unit circle $\Rightarrow$ the system is **non-minimum phase**.

Pole: $1-0.5z^{-1} = 0 \Rightarrow z = 0.5$, which lies *inside* the unit circle $\Rightarrow$ the system is **stable**.

Hence, the correct option is **(D)**.

Q6.12.1 MCQ 2012 2M Ans: C

Given $x[n] = \left(\dfrac{1}{3}\right)^{|n|} - \left(\dfrac{1}{2}\right)^{n} u[n]$.

The two-sided term splits as $\left(\frac{1}{3}\right)^{|n|} = \left(\frac{1}{3}\right)^{n} u[n] + \left(\frac{1}{3}\right)^{-n} u[-n-1]$, whose ROC is $\frac{1}{3} < |z| < 3$.

The term $-\left(\frac{1}{2}\right)^{n} u[n]$ is right-sided with ROC $|z| > \frac{1}{2}$.

The overall ROC is the intersection of the two:

$$\max\left(\frac{1}{3}, \frac{1}{2}\right) < |z| < 3 \implies \frac{1}{2} < |z| < 3.$$

Hence, the correct option is **(C)**.

Q6.11.1 MCQ 2011 2M Ans: B

Given $y(n) - \dfrac{1}{3}y(n-1) = x(n)$, $x(n) = \left(\dfrac{1}{2}\right)^{n} u(n)$. Taking the Z-transform under initial-rest conditions,

$$Y(z) - \frac{1}{3}z^{-1}Y(z) = \frac{z}{z-\frac{1}{2}} \Rightarrow Y(z) = \frac{z}{\left(z-\frac{1}{2}\right)\left(1-\frac{1}{3}z^{-1}\right)} = \frac{z^2}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{3}\right)}.$$

Expanding $Y(z)/z$ in partial fractions,

$$\frac{Y(z)}{z} = \frac{3}{z-\frac{1}{2}} - \frac{2}{z-\frac{1}{3}} \Rightarrow Y(z) = \frac{3z}{z-\frac{1}{2}} - \frac{2z}{z-\frac{1}{3}}.$$

Taking the inverse Z-transform,

$$y(n) = -2\left(\frac{1}{3}\right)^{n} + 3\left(\frac{1}{2}\right)^{n}, \quad n \geq 0.$$

Hence, the correct option is **(B)**.

Q6.10.1 MCQ 2010 2M Ans: B

For $H(z)$ to be stable, its poles must lie inside the unit circle. For the inverse system $1/H(z)$ to be stable, its poles — i.e., the *zeros* of $H(z)$ — must also lie inside the unit circle.
Hence both the poles *and* zeros of $H(z)$ must lie inside the unit circle, and the correct option is **(B)**.

Q6.08.1 MCQ 2008 1M Ans: A

Given $x(n) = 2^n u(n)$, a right-sided (causal) signal.

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} = \sum_{n=0}^{\infty} 2^n z^{-n} = \sum_{n=0}^{\infty} \left(2z^{-1}\right)^n.$$

This converges only if $|2z^{-1}| < 1$, i.e. $|z| > 2$, giving

$$X(z) = \frac{1}{1-2z^{-1}} = \frac{z}{z-2}, \qquad \text{ROC} : |z| > 2.$$

Hence, the correct option is **(A)**.

Q6.08.2 MCQ 2008 2M Ans: A

Given $y(n) = x(n) - \frac{1}{3}y(n-1)$, $x(n) = \delta(n)$, with $y(n) = 0$ for $n < 0$ (causal). Taking the Z-transform,

$$Y(z) = 1 - \frac{1}{3}z^{-1}Y(z) \;\Rightarrow\; Y(z)\left(1 + \frac{1}{3z}\right) = 1 \;\Rightarrow\; Y(z) = \frac{3z}{3z+1} = \frac{z}{z+\frac{1}{3}}.$$

Taking the inverse Z-transform,

$$y(n) = \left(-\frac{1}{3}\right)^n u(n).$$

Hence, the correct option is **(A)**.

Q6.05.1 MCQ 2005 2M Ans: D

From the given block diagram,

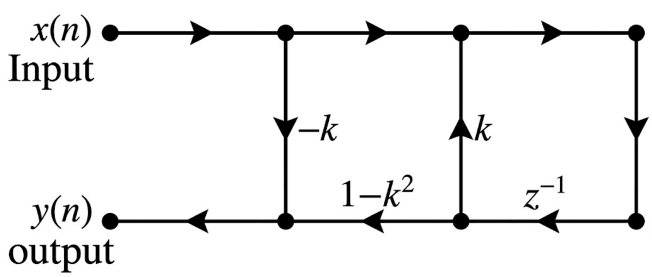

Fig. Solution Q6.05.1

$$W(z) = X(z) + z^{-1}\,k\,W(z) \;\Rightarrow\; W(z)\left(1 - kz^{-1}\right) = X(z) \;\Rightarrow\; W(z) = \frac{X(z)}{1-kz^{-1}}. \qquad \text{(i)}$$

Also,

$$Y(z) = -kX(z) + (1-k^2)z^{-1}W(z).$$

Substituting (i),

$$\frac{Y(z)}{X(z)} = -k + \frac{(1-k^2)z^{-1}}{1-kz^{-1}} = \frac{-k + k^2z^{-1} + z^{-1} - k^2z^{-1}}{1-kz^{-1}} = \frac{-k+z^{-1}}{1-kz^{-1}}.$$

Hence, the correct option is **(D)**.

Q6.04.1 MCQ 2004 2M Ans: C ● ● ○

Given:

$$X(z) = \frac{1/2}{1 - az^{-1}} + \frac{1/3}{1 - bz^{-1}} \quad \ldots \text{(i)}$$

where $|a| < 1$ and $|b| < 1$
ROC : $|a| < |z| < |b|$

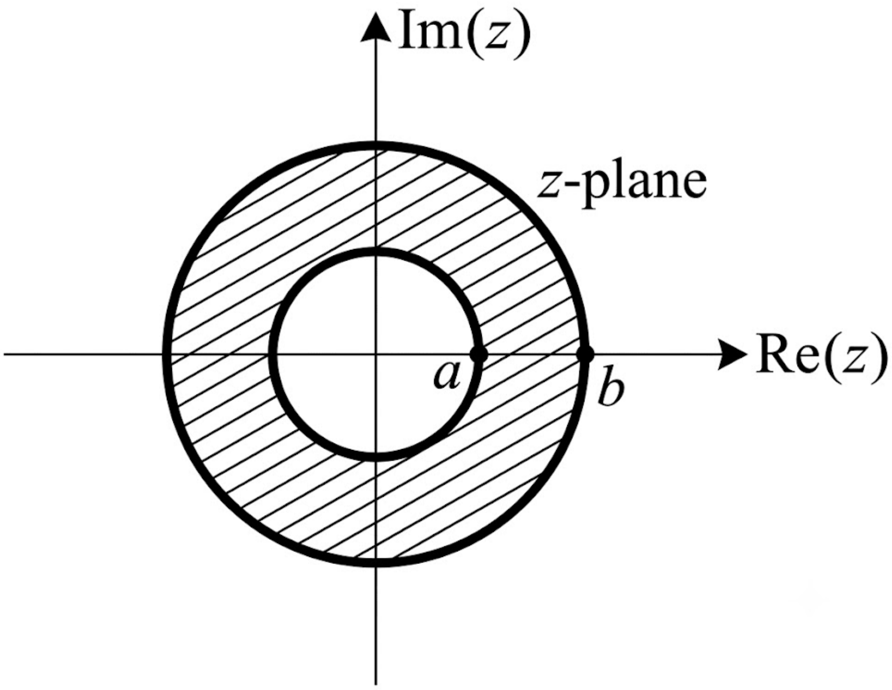

Fig. Solution Q6.04.1

According to given ROC,

$$X(z) = \underbrace{\frac{(1/2)z}{z - a}}_{\text{Right sided signal}} + \underbrace{\left(\frac{1}{3}\right)\frac{z}{z - b}}_{\text{Left sided signal}}$$

$$x(n) = \frac{1}{2}(a)^n u(n) - \frac{1}{3}(b)^n u(-n - 1)$$

$$x(0) = \frac{1}{2}$$

$$\boxed{\text{(C) } \frac{1}{2}}$$

Key Points

- An ROC of the form $|a| < |z| < |b|$ represents a two-sided sequence consisting of a right-sided part for $|z| > |a|$ and a left-sided part for $|z| < |b|$.
- The inverse z-transform yields $x(0) = \frac{1}{2}$ since $u(0) = 1$ and $u(-1) = 0$.

Q6.04.2 MCQ 2004 1M Ans: A

Given $H(z) = \dfrac{1}{1 - 0.7z^{-1} + 0.13z^{-2}}$, realized in direct form with feed-forward gain a_0 and feedback gains a_1, a_2 as shown in the figure.

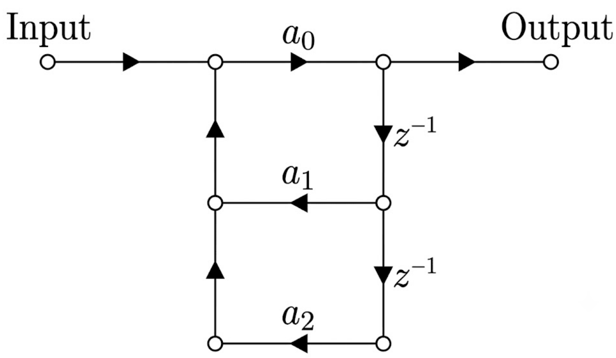

Fig. Solution Q6.04.2

Comparing the denominator coefficients of $H(z)$ with the direct-form structure,

$$a_0 = 1.0, \qquad a_1 = 0.7, \qquad a_2 = -0.13.$$

Hence, the correct option is **(A)**.

Q6.04.3 MCQ 2004 2M Ans: D

Given: Discrete time sequence $x(n)$ suffered distortion modeled by an LTI system with,

$$H(z) = (1 - az^{-1}); \quad |a| > 1$$

To compensate the distortion, the system transfer function of a stable system should be $\frac{1}{H(z)}$.

$$\frac{1}{H(z)} = \frac{1}{1 - az^{-1}}$$

Inverse transform is given by,
For right sided signal,

$$h_c(n) = a^n u(n); \quad \text{ROC} : |z| > |a| \quad \ldots\text{(i)}$$

For left sided signal,

$$h_c(n) = -a^n u(-n-1); \quad \text{ROC} : |z| < |a| \quad \ldots\text{(ii)}$$

For a stable system, the ROC must include the unit circle ($|z| = 1$). Since $|a| > 1$, the ROC $|z| < |a|$ includes the unit circle, so equation (ii) represents the stable impulse response.

$$\boxed{\text{(D)} \; -a^n u(-n-1)}$$

Key Points

- An LTI system is BIBO stable if and only if the Region of Convergence (ROC) of its system function $H(z)$ contains the unit circle ($|z| = 1$).
- When a pole lies outside the unit circle ($|a| > 1$), a stable system requires a left-sided impulse response with ROC $|z| < |a|$.

Q6.03.1 MCQ 2003 1M Ans: A

Given $X(z) = e^{1/z}$. Expanding as a power series in z^{-1},

$$X(z) = e^{1/z} = 1 + \frac{z^{-1}}{1!} + \frac{z^{-2}}{2!} + \frac{z^{-3}}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n}. \quad \text{(i)}$$

By definition,

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}. \quad \text{(ii)}$$

Comparing (i) and (ii):

$$x(n) = \frac{1}{n!} \text{ for } n \geq 0 \quad \Rightarrow \quad x(n) = \frac{u(n)}{n!}.$$

Hence, the correct option is **(A)**.

Q6.03.2 MCQ 2003 2M Ans: A

Given input $x = [a, b, c, d]$ and output $y = [x, x, x, x, \ldots \langle\text{repeated } N \text{ times}\rangle]$, i.e.

$$x(n) = a\delta(n) + b\delta(n-1) + c\delta(n-2) + d\delta(n-3),$$

$$y(n) = a\delta(n) + b\delta(n-1) + c\delta(n-2) + d\delta(n-3) + a\delta(n-4) + b\delta(n-5) + \cdots$$

The output repeats after every 4th sample, so the fundamental period is $N = 4$, and

$$y(n) = \sum_{i=0}^{N-1} x(n-4i) \quad \Longrightarrow \quad Y(z) = \sum_{i=0}^{N-1} X(z) z^{-4i}.$$

$$H(z) = \frac{Y(z)}{X(z)} = \sum_{i=0}^{N-1} z^{-4i} = 1 + z^{-4} + z^{-8} + \cdots + z^{-4(N-1)},$$

$$h(n) = \sum_{i=0}^{N-1} \delta(n-4i).$$

Per the official Answer Key, the correct option is **(D)**.

Q6.01.1 MCQ 2001 1M Ans: A

Given $x(n) = \delta(n) \xrightarrow{h(n)} y(n)$. By the convolution property, $y(n) = x(n) \otimes h(n) \xrightarrow{\text{Z.T.}} Y(z) = X(z)H(z)$. With $x(n) = \delta(n) \leftrightarrow 1$ and $h(n) = a^n u(n) \leftrightarrow \frac{z}{z-a}$,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z}{z-a}.$$

For a unit-step input $x_1(n) = u(n) \leftrightarrow \frac{z}{z-1}$, ROC : $|z| > 1$.

$$Y_1(z) = \frac{z}{z-1} \cdot \frac{z}{z-a} = \frac{1}{1-a} \cdot \frac{z}{z-1} + \frac{a}{a-1} \cdot \frac{z}{z-a}.$$

Taking the inverse Z-transform,

$$y(n) = \frac{1}{1-a} u(n) + \frac{a}{a-1} a^n u(n) = \sum_{j=0}^{n} a^j.$$

Hence, the correct option is **(A)**.